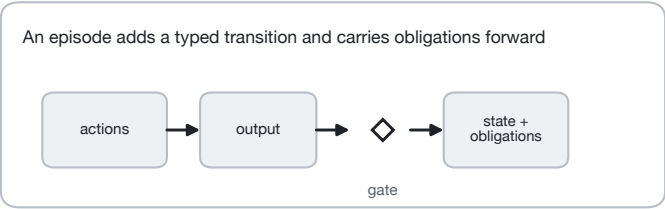
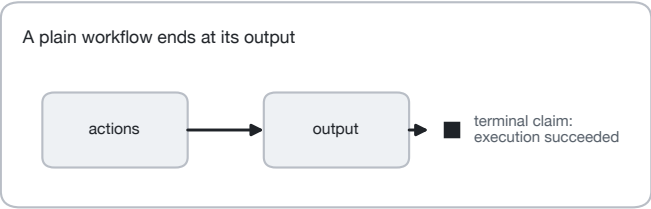
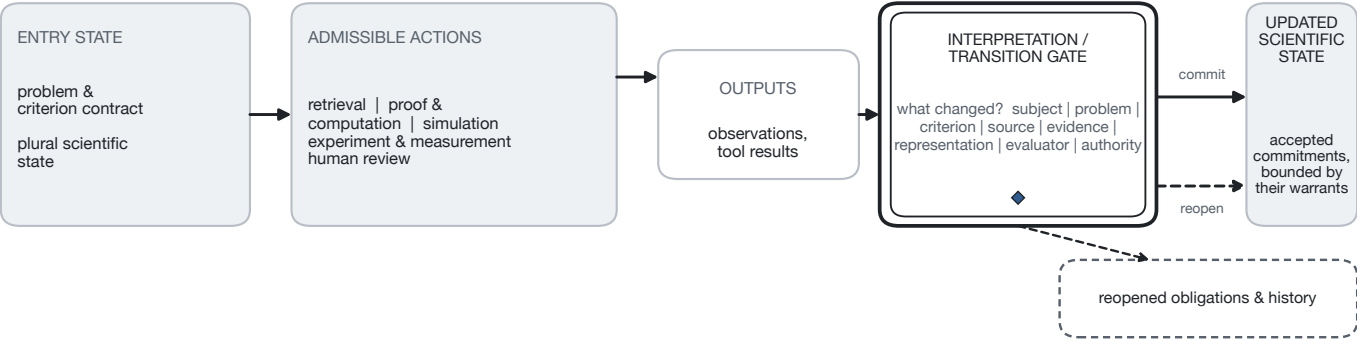


Executable research episodes change scientific state, not only outputs

Inside one machine-mediated research episode every action passes through a typed interpretation/transition gate before any scientific commitment changes.



Four failures of transition that local success does not detect

1

Execution succeeds - evidence link invalid

✗

genuine papers retrieved; claim bound to the wrong passage

2

Provenance & replay exact - analysis wrong

✗

the record is complete; lineage does not establish scientific correctness

3

Three validators agree - one hidden source

✗

apparent independent agreement is dependent; count is not independence

4

Local search flat - a route is censored

✗

absence of evidence on searched routes is not evidence of closure

Panels 1-4 are constructed test cases with known correct answers, not prevalence estimates.

Published anchors: provenance does not establish correctness [35, 36, 54]; dependent agreement [37]; consensus is not truth [40, 41]; censored routes [65].